

# Project Design Phase-II

## Data Flow Diagram & User Stories

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Team ID: LTVIP2026TMIDS72781

Project Name: Flight Booking App (MERN Stack)

Maximum Marks: 4 Marks

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### Data Flow Diagram (DFD) - Level 0

- External Entities: Customer (Mobile/Web User), Administrator, Payment Gateway.
- Main Process: Flight Booking System.
- Data Stores: User Database, Flight Database, Booking Database.
- Customer sends registration/login data to system and receives authentication status.
- Customer searches flights and system returns available flight list.
- Customer submits booking details and system processes payment via payment gateway.
- System stores booking details and sends booking confirmation to customer.
- Administrator manages flight records (Add/Edit/Delete).

### User Stories

|               | Epic              | USN   | User Story                                   | Acceptance Criteria                 | Priority |
|---------------|-------------------|-------|----------------------------------------------|-------------------------------------|----------|
| (Mobile)      | Registration      | USN-1 | Register using email & password              | Access dashboard after registration | High     |
| (Mobile)      | Login             | USN-2 | Login using email & password                 | Redirect to dashboard               | High     |
| (Web)         | Flight Search     | USN-3 | Search flights by source, destination & date | Correct flight list displayed       | High     |
| (Web)         | Booking           | USN-4 | Book selected flight                         | Booking confirmation generated      | High     |
| (Web)         | Booking History   | USN-5 | View past bookings                           | Booking history displayed           | Medium   |
| Administrator | Flight Management | USN-6 | Add new flights                              | Flight visible in search results    | High     |
| Administrator | Flight Management | USN-7 | Edit/Delete flights                          | Changes reflected immediately       | High     |
|               | Payment           | USN-8 | Complete secure payment                      | Payment success message shown       | High     |